

```
CREATE DATABASE Pwskills_ETL;
USE Pwskills_ETL;
CREATE TABLE customers (
    Customer_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    City VARCHAR(50),
    Monthly_Sales INT,
    Income INT,
    Region VARCHAR(20)
);
```

```
INSERT INTO customers VALUES
(101, 'Rahul Mehta', 'Mumbai', 12000, 65000, 'West'),
(104, 'Neha Singh', 'Delhi', NULL, NULL, 'North'),
(102, 'Anjali Rao', 'Bengaluru', NULL, NULL, 'South'),
(105, 'Amit Verma', 'Pune', 18000, 58000, NULL),
(107, 'Pooja Das', 'Kolkata', 14000, NULL, 'East'),
(103, 'Suresh Iyer', 'Chennai', 15000, 72000, 'South'),
(106, 'Karan Shah', 'Ahmedabad', NULL, 61000, 'West'),
(108, 'Riya Kapoor', 'Jaipur', 16000, 69000, 'North');
```

question 8

```
SELECT *
FROM customers
WHERE Region IS NOT NULL;
SELECT COUNT(*) AS Records_Lost
FROM customers
WHERE Region IS NULL;
```

question 9

```
SELECT
    Customer_ID,
    Name,
    City,
    Monthly_Sales,
    COALESCE(
        Monthly_Sales,
        MAX(Monthly_Sales) OVER (
            ORDER BY Customer_ID
            ROWS BETWEEN UNBOUNDED PRECEDING AND 1 PRECEDING
        )
    )
```

```
)  
) AS Forward_Filled_Sales,  
Income,  
Region  
FROM customers;
```

question 10

```
SELECT  
    Customer_ID,  
    Name,  
    City,  
    Monthly_Sales,  
    Income,  
    Region,  
    CASE  
        WHEN Income IS NULL THEN 1  
        ELSE 0  
    END AS Income_Missing_Flag  
FROM customers;
```

Q1. Most common reasons for missing data in ETL pipelines

Data entry errors

Optional fields left blank

System / sensor failures

Integration issues between sources

Corrupted or lost records

Different schema across systems

Q2. Why blindly deleting rows is bad

Causes data loss

Reduces sample size

Introduces bias

May remove important patterns

Distorts analytics & ML models

Q3. Listwise vs Column Deletion

Listwise Deletion

→ Remove rows with missing values

✓ Appropriate when:

Very few rows affected

Missing data is random

Column Deletion

→ Remove entire column

✓ Appropriate when:

Column has too many missing values

Column not critical for analysis

Q4. Median vs Mean (skewed data)

Median is preferred because:

Resistant to outliers

Represents central tendency better

Mean gets distorted in skewed distributions (like income)

Q5. Forward Fill

Forward Fill = Replace missing values with previous valid value

✓ Useful in:

Time-series data

Sequential records

Sales / stock / sensor datasets

Q6. Why flag missing before imputation

Preserves missingness information

Missing data itself may hold patterns

Helps ML models detect anomalies

Enables business insights

Q7. Missing income → Business insights

Missing income may indicate:

Customers unwilling to disclose income

Certain demographic segment

Data collection issue

Privacy-sensitive customers

Potential high/low income group pattern

SECTION B – PRACTICAL

Given Dataset (Original)

ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
104	Neha Singh	Delhi	NaN	NaN	North
102	Anjali Rao	Bengaluru	NaN	NaN	South
105	Amit Verma	Pune	18000	58000	NaN
107	Pooja Das	Kolkata	14000	NaN	East
103	Suresh Iyer	Chennai	15000	72000	South
106	Karan Shah	Ahmedabad	NaN	61000	West
108	Riya Kapoor	Jaipur	16000	69000	North

Q8. Listwise Deletion (Remove missing Region)

Affected Row

Customer 105 (Amit Verma)

Dataset After Deletion

ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
104	Neha Singh	Delhi	NaN	NaN	North
102	Anjali Rao	Bengaluru	NaN	NaN	South
107	Pooja Das	Kolkata	14000	NaN	East
103	Suresh Iyer	Chennai	15000	72000	South
106	Karan Shah	Ahmedabad	NaN	61000	West
108	Riya Kapoor	Jaipur	16000	69000	North

Records Lost: 1

Q9. Imputation – Forward Fill (Monthly_Sales)

Before → After

ID	Name	Before	After
104	Neha Singh	NaN	12000
102	Anjali Rao	NaN	12000
106	Karan Shah	NaN	15000

Explanation

Forward fill suitable because:

Data appears sequential

Sales often follow time/order continuity

Prevents unrealistic averages

Q10. Flagging Missing Income
Income_Missing_Flag

(0 = Present, 1 = Missing)

ID	Name	Income	Flag
101	Rahul Mehta	65000	0
104	Neha Singh	NaN	1
102	Anjali Rao	NaN	1
105	Amit Verma	58000	0
107	Pooja Das	NaN	1
103	Suresh Iyer	72000	0
106	Karan Shah	61000	0
108			